EXPT NO:3	CLEANSE DATA BY HANDLING MISSING VALUES, OUTLIERS, AND INCONSISTENCIES USING TABLEAU AND POWERBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What are the potential impacts of missing values in a dataset, and what are some common strategies to address them in Tableau and Power BI?
   Missing values can reduce the accuracy of data analysis.
   They may cause incorrect calculations and misleading visualizations.
   Numeric nulls are replaced using mean or median values.
   Text nulls are replaced with values like "Unknown".
   Tableau and Power BI provide built-in tools and functions to handle missing values.

2. What is the importance of identifying and addressing outliers in a dataset? How can outliers affect data analysis and model performance?
   Outliers are values that differ significantly from other data points.
   They can distort averages and overall trends.
   Outliers may reduce the accuracy of predictive models.
   They should be filtered, capped, or reviewed carefully.
   Handling outliers improves the reliability of analysis.

3. What types of data inconsistencies might you encounter during data cleansing, and how can Tableau and Power BI help in resolving them?
   Data inconsistencies include spelling errors and duplicate values.
   Different date or currency formats may also occur.
   Tableau provides Group and Alias features to standardize data.
   Power BI uses Replace Values and Power Query transformations.
   These tools improve data consistency and quality.

4. How do you identify patterns in data using visualizations to spot missing values, outliers, and inconsistencies?
   Charts help identify unusual patterns in data quickly.
   Scatter plots and box plots reveal outliers clearly.
   Bar charts highlight missing or inconsistent categories.
   Histograms show the distribution of numerical values.
   These visuals simplify the data cleansing process.

5. What is the role of calculated fields in data cleansing, and how can they be used to handle missing values and outliers in Tableau/Power BI?
   Calculated fields automate common data cleaning tasks.
   They replace missing values using logical conditions.
   They can identify and flag outliers for review.
   Business rules can be applied consistently using formulas.
   This improves the accuracy and reliability of the dataset.

IN-LAB EXERCISE:

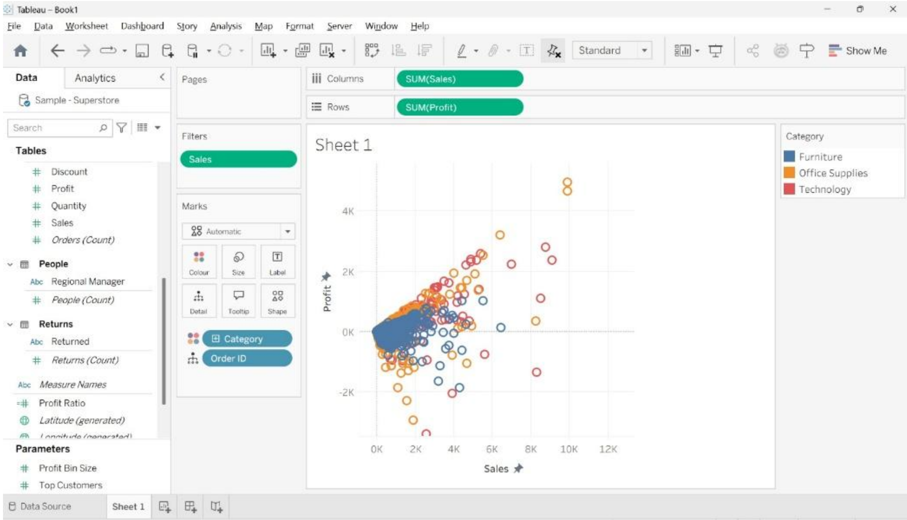
OBJECTIVE:

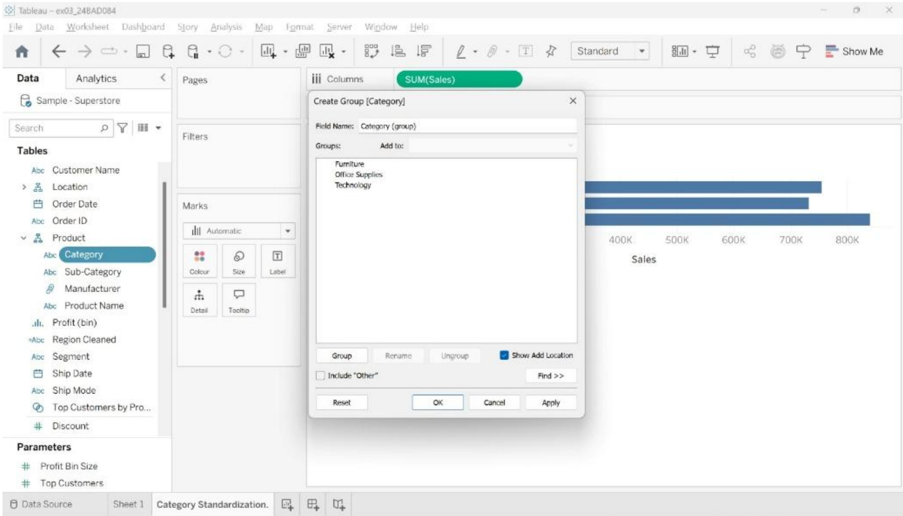
To clean a given dataset (e.g., Superstore Sample in Tableau and Financials/Sales Data in Power BI) by identifying and resolving missing values, outliers, and inconsistencies using built-in tools, ensuring the dataset is accurate and ready for analysis.

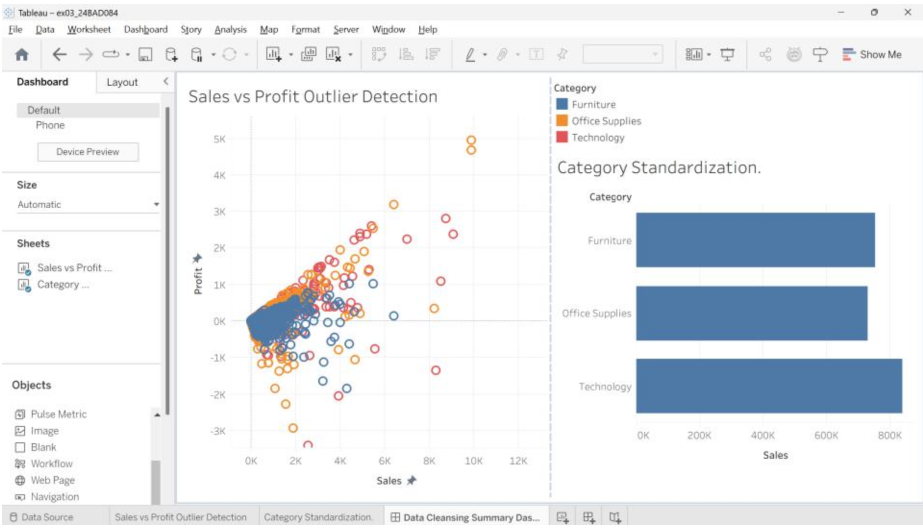
PROCEDURE:

1. **Open the Tool and Connect Dataset** ○ Open Tableau Desktop , Power BI Desktop.
    - ○ Load the built-in dataset (Sample - Superstore in Tableau / Financials in Power BI).
    - ○ Preview the dataset to understand its structure.
2. **Initial Inspection** ○ Explore fields such as Sales, Profit, Region, or Product Category.
    - ○ Check for nulls, inconsistent text values (e.g., "South Region" vs. "South"), or date formatting issues.
3. **Handling Missing Values** ○ Use Data Interpreter (Tableau) / Power Query (Power BI) to highlight nulls. ○ Replace missing numerical values with mean/median, and text with "Unknown".
    - ○ Use calculated fields: IFNULL([Field], Default) in Tableau or IF(ISBLANK([Field]), "Unknown", [Field]) in Power BI.
4. **Detecting and Managing Outliers** ○ Create scatter or box plots for Sales and Profit to identify extreme values.
    - ○ Choose strategies:
        - Filter them temporarily.
        - Apply caps using calculated columns.
        - Flag them with labels for review.
5. **Resolving Inconsistencies** ○ Use Group/Alias (Tableau) / Replace Values (Power BI) to standardize text.
    - ○ Normalize data formats, such as date fields or currency units.
6. **Validation with Visuals**
    - ○ Build dashboards with histograms, bar charts, and box plots to visually confirm improvements.
    - ○ Compare before/after cleaning stages for accuracy.
7. **Save the Cleaned Dataset** ○ Save as .twbx (Tableau) / .pbix (Power BI).
    - ○ Optionally export cleaned data to .csv.
8. **Summary Dashboard**
    - ○ Create a visual summary of the cleaning process, highlighting:
        - Fields corrected.
        - Nulls/outliers removed or adjusted.

- Clean dataset metrics (record count, completeness %).







Experiment Dataset:

1. Tableau: *Sample - Superstore*
2. Power BI: *Financials* or *Sales Data*

How to Access:

Tableau:

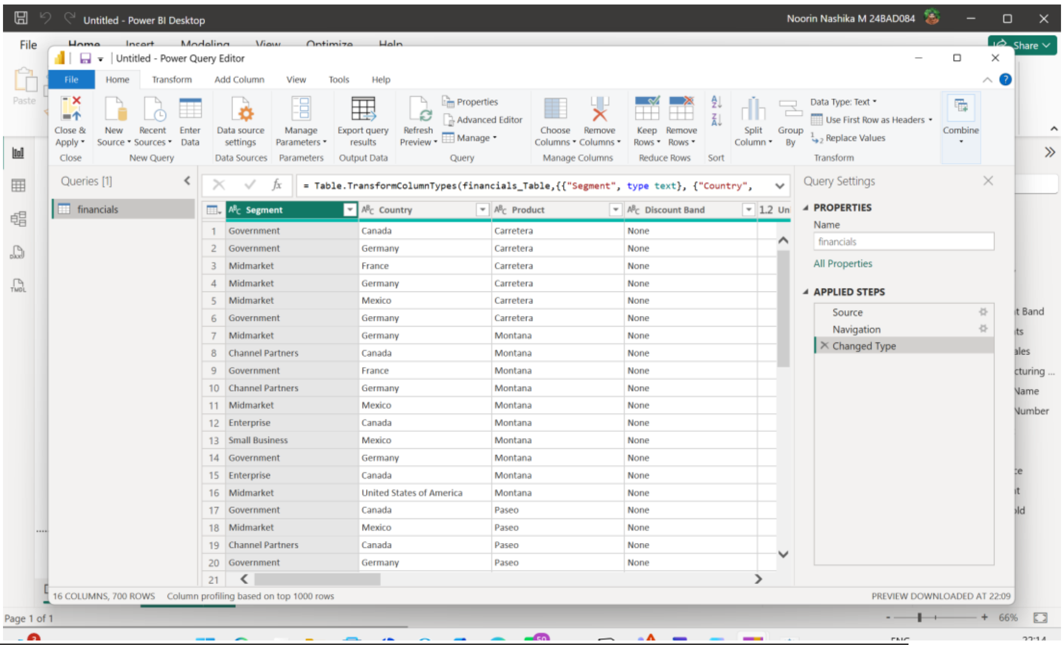
- **Navigate to:** *Start Page → Sample Workbooks*
- Or: *Help → Sample Data → Open Sample Data Folder*

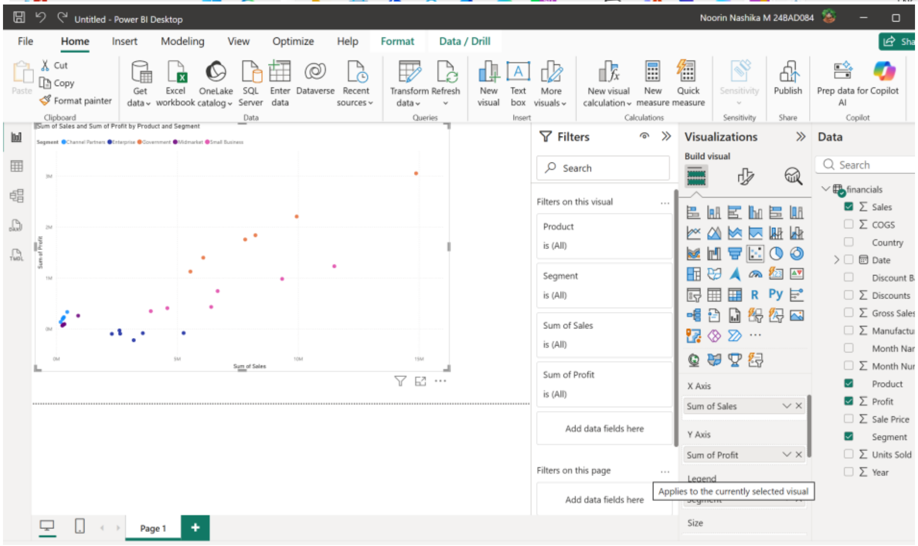
Power BI:

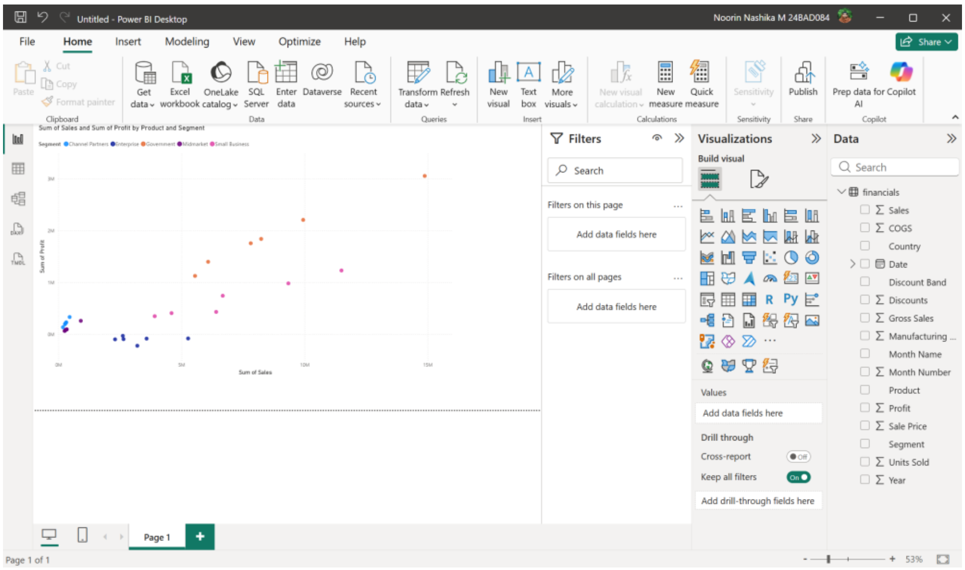
- **Navigate to:** *Home → Get Data → Excel → Choose sample datasets (e.g., Financial Sample.xlsx)*
- Microsoft also hosts Power BI sample datasets online.

Fields Used:

1. Sales
2. Profit
3. Region
4. Category
5. Order Date / Transaction Date







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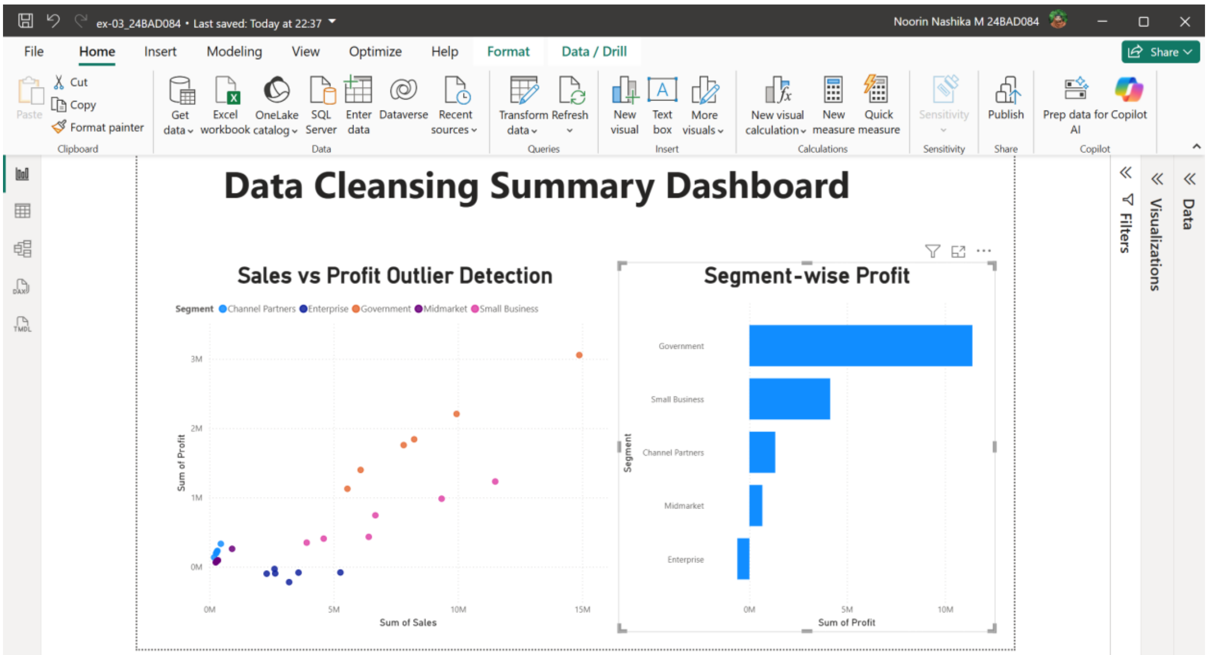

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What method did you use to replace missing values in numeric fields?
   Missing numeric values were identified during data inspection.
   The mean or median value was used for replacement.
   Text fields were filled with "Unknown" where necessary.
   Tableau used IFNULL(), while Power BI used IF(ISBLANK()).
   This ensured complete and consistent data for analysis.

2. How were outliers handled in your dataset and what visual confirmed their correction?
   Outliers were detected using scatter plots and box plots.
   Extreme values were reviewed before making changes.
   They were filtered or capped using calculated fields.
   Updated visualizations showed a more balanced distribution.
   This confirmed that the outliers were handled successfully.

3. What inconsistencies were detected in categorical values and how were they standardized?
   Different spellings and naming formats were identified.
   Duplicate category names were also checked.
   Tableau's Group and Alias features standardized the values.
   Power BI used Replace Values in Power Query.
   This ensured uniform categorical data throughout the dataset.

4. Describe how calculated fields helped in resolving data issues.
   Calculated fields replaced null values automatically.
   They applied logical conditions to clean the data.
   Outliers were identified using conditional expressions.
   Repeated calculations became easier and more consistent.
   This improved overall data quality and analysis.

5. How did your dashboard demonstrate the effectiveness of the data cleaning process?

The dashboard displayed cleaner and more consistent visuals.
Missing values and inconsistencies were no longer visible.
Charts showed accurate trends and distributions.
The cleaned data produced reliable insights for analysis.
This confirmed the success of the data cleansing process.

RESULT:

The dataset was successfully cleaned and prepared for accurate analysis using Tableau and Power BI. Issues such as nulls, inconsistencies, and outliers were identified and resolved, improving data quality significantly.

ASSESSMENT

Description	**Max Marks**	**Marks Awarded**
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
Faculty Signature		